



Computer (& AI) Methods for Structural Engineering (DISC 5203)

Fall 2026

Homework 1 - Due date: 22/9/2026 (1st) and 29/9/2026 (final)

Homework Assignment 1: AI-Assisted Structural Analysis of a Building Floor Plan

Context

This assignment replaces a conventional midterm. It has two purposes: to apply structural engineering fundamentals, including tributary areas and axial loads, and to gain hands-on experience with AI agent-based coding tools in a real project workflow.

The course strongly encourages the use of an AI agent tool rather than a plain chatbot. The agent should read files, write scripts, run code, check results, and iterate autonomously within the project folder.

Input

A building floor plan image is provided according to your student ID number.

Assume that the supplied floor plan represents a typical floor of a **5-storey building**. The same floor plan is used on all five floors and assume the height of all floors is 3m. Also assume a live area load of $5kN/m^2$ and assume a slab thickness of $h = 150mm$ (if not specified in the floor plan) to calculate the dead area load.

Tasks

Using an AI agent, work through the following steps and document your methodology:

1. Extract the floor outline from the image and compute the total floor area.
2. Identify the vertical structural elements (columns, walls, cores) and report their coordinates or locations.
3. Determine the tributary area assigned to each vertical element.
4. Calculate the dead load, live load, and total axial load carried by each vertical element at its level.
5. Produce a summary table for every vertical element containing its ID, location, tributary area, dead load, live load, and total axial load.

There is no unique way to extract information such as the coordinates of vertical elements from an image file. Propose, implement, and document your own defensible solution. You may want to explore readily available, general-purpose algorithms for bisecting the distances between points in two dimensions when determining tributary areas.

Deliverables

Submit a Markdown (.md) report containing:

- A description of your image-reading and calculation methodology.
- The extracted floor outline, element coordinates or locations, assumptions, and results.
- Tables showing tributary areas and dead, live, and total axial loads.
- Any plots or diagrams generated by the agent, embedded in the Markdown report.
- Units, equations, intermediate values, and verification checks sufficient to reproduce and defend the results.

Also submit the project repository managed with Git, including scripts, configuration files (CLAUDE.md or equivalent guidance file, skills.md, and any hooks), and intermediate outputs.

Required Tools & Workflow

- Use an **AI agent tool** (Claude Code, OpenCode, DeepSeek Harness, or an equivalent harness) that can read/write files, execute scripts, and iterate on results within a project environment.
- Set up a project structure that may include a CLAUDE.md (or equivalent) guidance file, a skills.md for reusable procedures, and hooks for deterministic actions.
- Use **Git** to track changes and maintain a safety net for your work.
- Do not rely solely on a conversational chatbot. Use the agent's ability to run code, inspect outputs, and revise iteratively.
- Review the agent's code and results yourself. Verify calculations, units, image interpretation, and structural logic.

Suggested Workflow

1. Open a project folder and initialize Git before beginning.
2. Let the agent inspect the floor-plan image and create extraction scripts (you need a visual LLM).
3. Build the calculation workflow step by step: floor outline, vertical elements, tributary areas, and loads.
4. Ask the agent to plot the extracted geometry and assemble the Markdown report.
5. Review the results and use `git diff` to inspect meaningful changes.
6. Commit to Git after each meaningful step.

Rules & Expectations

- **Individual work.** The assignment must be completed individually so that you fully understand every step of the process.
- **You must be able to defend your submitted work.** Be prepared to explain your methodology, justify assumptions, and answer questions about both the engineering content and the AI tooling workflow.
- **Verify all AI outputs.** AI models can hallucinate or make errors with high confidence. You are responsible for checking calculations, unit consistency, and structural logic.
- The **known input** (the provided floor plan) is the major component of the grade.
- An **unknown input**, such as a different or more complex floor plan, will account for approximately 20% of the grade. Your methodology should therefore generalize beyond one image.

Deadline & Checkpoint

Milestone	Due
Checkpoint - partial results and evidence of using the tools	22/9/2026 (Week 4)
Final submission - full Markdown report and Git repository	29/9/2026 (Week 5)

The course is currently in Week 3. At the checkpoint, submit evidence that you have used the tools and produced intermediate results, even if the complete solution is not finished.

What to Submit

1. The final **Markdown report** containing your methodology, results, calculations, and explanations.
2. The **project repository** (Git) with your code, configuration files (CLAUDE.md, skills.md, hooks, etc.), supporting scripts, plots, and intermediate outputs.
3. Be prepared to **demonstrate and discuss** your workflow in a follow-up session.

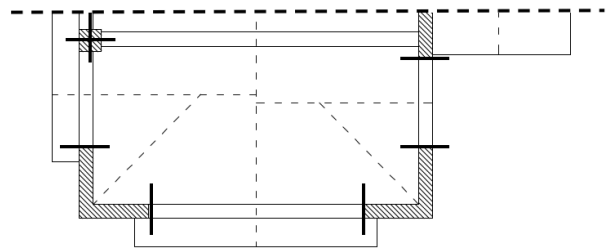
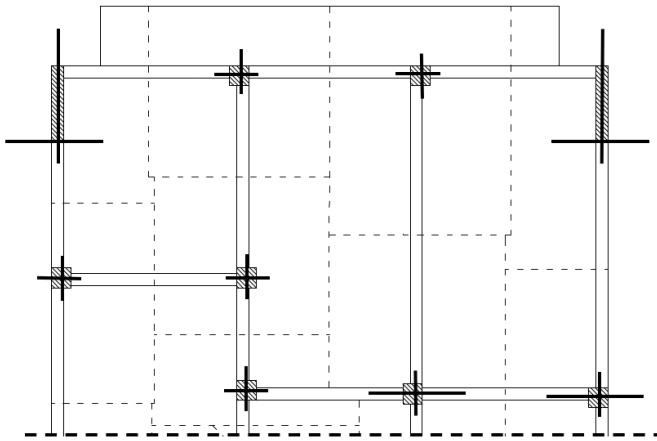
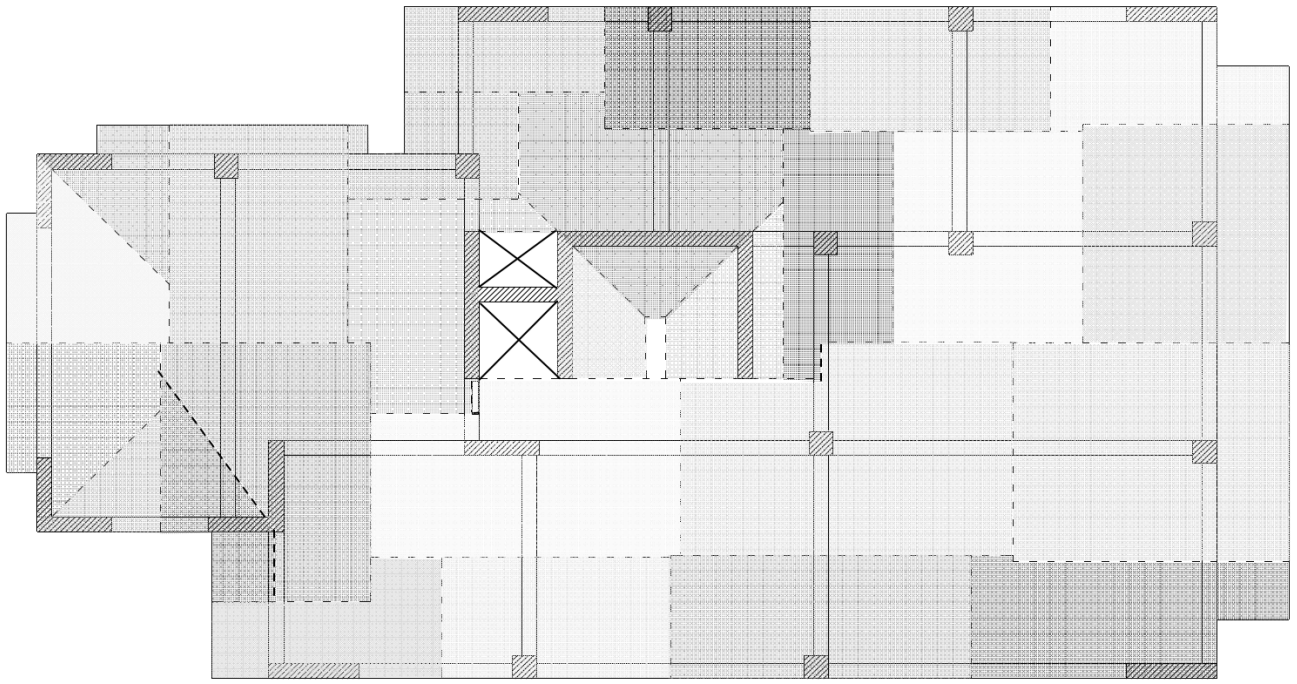
What Comes Next

The axial-load results from this assignment will feed into later work on beam/slab equivalent loads and a more complete picture of how a multi-storey building transfers gravity loads to its vertical elements.

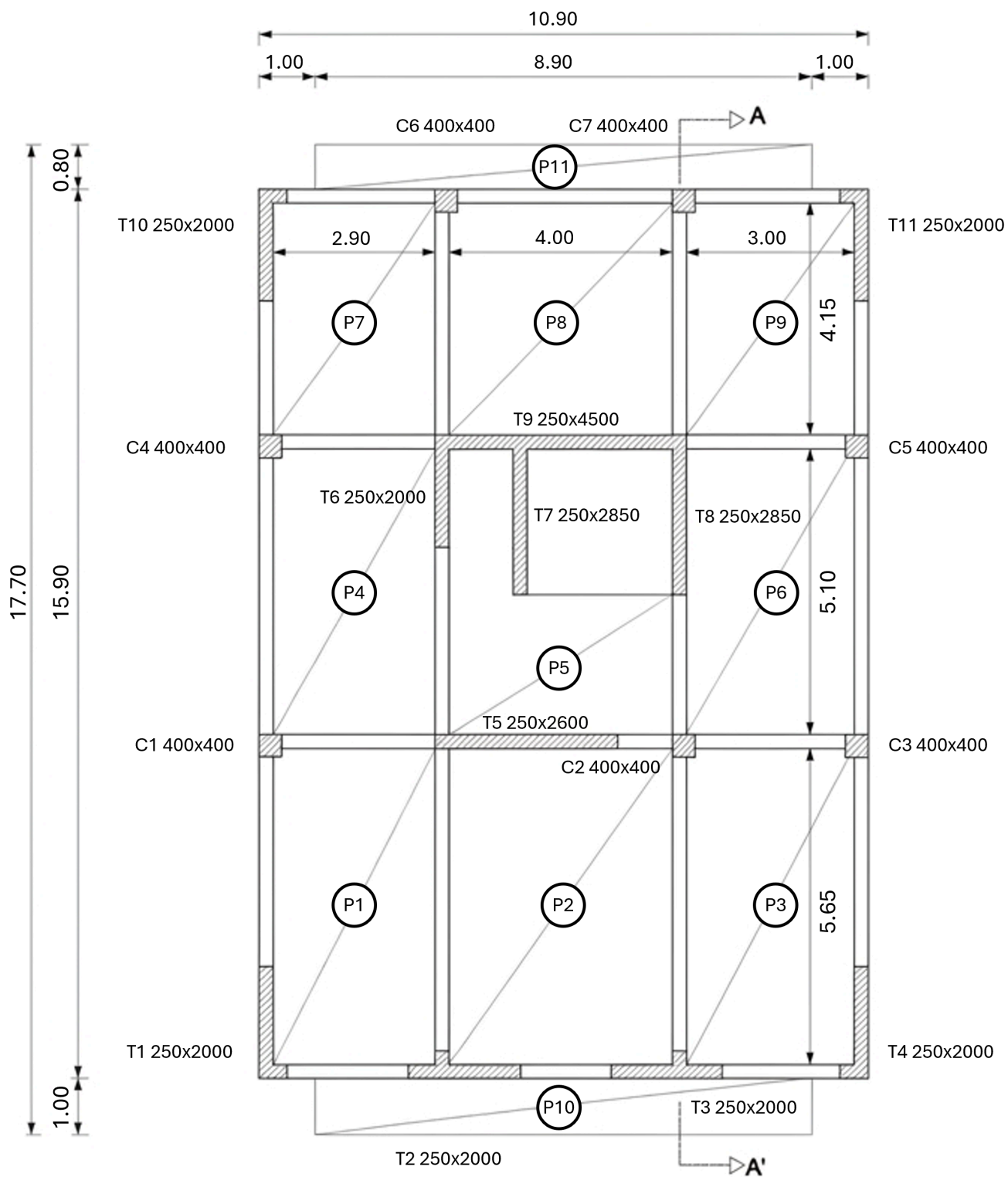
Note: The floor plan image and the VS Code environment walkthrough will be provided in the next session.

hint: Rules for Defining Tributary (Influence) Areas

Step	Rule	Practical Execution
1	Establish Local Support Axes	• Columns: Place support axes directly along the centerlines of the columns. • Shear Walls (Major/Span Direction): Place the local axis along the designated fixation line (green line). • Shear Walls (Minor/Transverse Direction): Place the axis of symmetry along the wall's centerline (red line).
2	Delineate Mid-Span Boundaries	• Calculate the mid-span distance between the established axes of adjacent vertical supports. • Draw boundary lines (blue dashed lines) at these midpoints to divide the floor plan into tributary sections.
3	Construct Influence Surfaces	• Repeat the axis and midpoint steps for every vertical element across the layout. • The closed areas formed by these boundary lines define the unique tributary (influence) surface area for each column and wall.
4	Resolve Corner Geometry	• Where perpendicular corner walls meet, project the boundary line outward from the corner vertex at a 45° angle until it intersects the regular mid-span boundary lines.

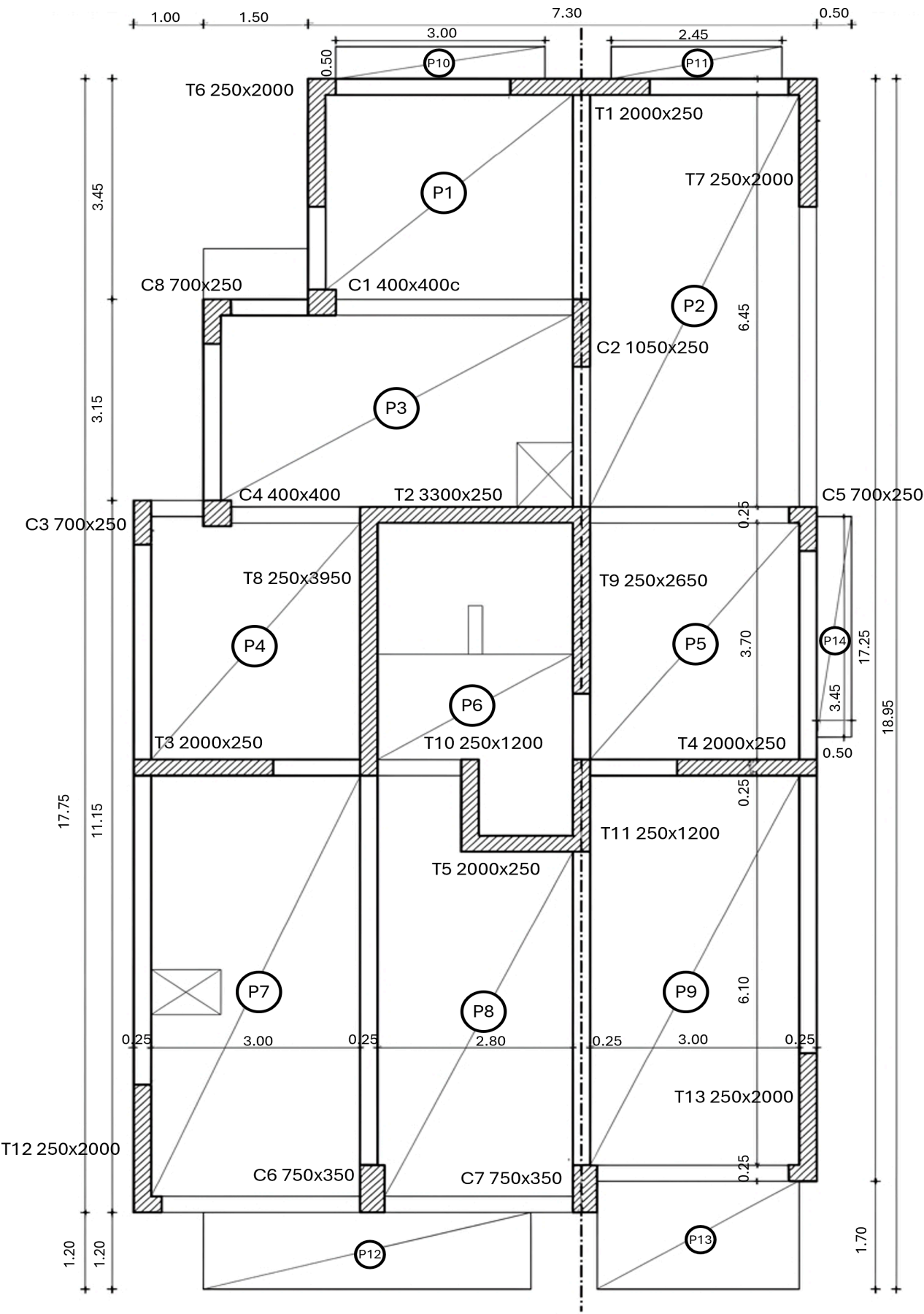


Floor Plans 1 (last digit of your student ID is 1, 2, or 3)



h= 15cm

Floor Plans 2 (last digit of your student ID is 4, 5, or 6)



Floor Plans 3 (last digit of your student ID is 7, 8, 9, or 0)

